

Reference Papers Summary Table

April 2026

1 Main Papers Summary

#	Title	Year	Algorithm Used	Paper Type
01	Service Composition in Dynamic Environments: A Systematic Review and Future Directions	2020	Multiple (GA, PSO, ACO, etc.)	Systematic Review
02	An approach based on genetic algorithms and neural networks for QoS-aware IoT services composition	2022	Genetic Algorithm + Neural Network (Hybrid)	Original Research
03	A Genetic Algorithm Based Approach for Fluctuating QoS Aware Selection of IoT Services	2020	Genetic Algorithm	Original Research
04	Spatio-Cohesive Service Selection Using Machine Learning in Dynamic IoT Environments	2020	Genetic Algorithm + Machine Learning (Hybrid)	Original Research
05	Service selection mechanisms in the Internet of Things (IoT): a systematic and comprehensive study	2020	Multiple (GA, PSO, ACO, ML, MCDM)	Systematic Review
06	A deep reinforcement learning based multiple meta-heuristic methods approach for resource constrained multi-project scheduling problem	2021	Deep Reinforcement Learning (DQN) + GA + PSO + SA + ABC (Hybrid)	Original Research

#	Title	Year	Algorithm Used	Paper Type
07	DDAPSO: Hybrid Discrete Dragonfly Algorithm and Particle Swarm Algorithm to Service Selection and Composition for the Internet of Things Applications	2022	Discrete Dragonfly Algorithm + PSO + Levy Flight (Hybrid)	Original Research
08	Reinforcement Learning for Interactive QoS-Aware Services Composition	2022	Q-Learning (Reinforcement Learning)	Original Research
09	Energy Centered and QoS Aware Services Selection for Internet of Things	2022	Genetic Algorithm (Multi-Objective)	Original Research
10	Incentive Based Selection and Composition of IoT Energy Services	2022	Game Theory + Optimization	Original Research
11	Machine Learning in Real-Time Internet of Things (IoT) Systems: A Survey	2022	Multiple (SL, UL, RL, DL, FL)	Survey
12	Composing Energy Services in a Crowdsourced IoT Environment	2022	Crowdsourcing + Optimization	Original Research
13	Crowdsourcing Energy as a Service	2022	CEaaS Framework + Market Mechanism	Original Research
14	Fluid Composition of Intermittent IoT Energy Services	2022	Fluid Composition + Adaptation Algorithm	Original Research
15	Proactive Composition of Mobile IoT Energy Services	2022	Trajectory Prediction + Proactive Planning	Original Research
16	Elastic Composition of Crowdsourced IoT Energy Services	2022	Genetic Algorithm + Elastic Adaptation	Original Research
17	A Deep Reinforcement Learning Approach for Composing Moving IoT Services	2022	Double DQN (Deep Reinforcement Learning)	Original Research

2 Summary Statistics

Metric	Value
Total Papers	17
Systematic Re-views/Surveys	3
Original Research	10
Year Range	2020-2022
Papers using Genetic Algorithm	6
Papers using Machine Learning/RL	6
Papers using Hybrid Approaches	6
Papers using Swarm Intelligence	3
Papers on Energy Efficiency	2

3 Algorithm Distribution

Algorithm	Papers Using
Genetic Algorithm (GA)	02, 03, 04, 09, 16
Neural Networks (NN)	02, 04
Particle Swarm Optimization (PSO)	01, 05, 06, 07
Ant Colony Optimization (ACO)	01, 05
Dragonfly Algorithm (DFA)	07
Machine Learning (General)	04, 05
Reinforcement Learning (RL)	08, 17
Q-Learning	08
Deep Reinforcement Learning (DQN)	06, 17
Double DQN	17

Algorithm	Papers Using
Multi-Criteria Decision Making (MCDM)	05
Simulated Annealing (SA)	06
Artificial Bee Colony (ABC)	06
Levy Flight	07
Opposition-Based Learning	07
Multi-Objective Optimization	09
Game Theory	10
Supervised Learning (SL)	11
Unsupervised Learning (UL)	11
Federated Learning (FL)	11
Elastic Adaptation	16

4 Datasets Used

Paper	Dataset	Description
01	147 Primary Studies	Systematic review - published papers from IEEE Xplore, ACM DL, Scopus, Web of Science
02	Synthetic IoT Services	Generated IoT service registry with QoS attributes
03	Synthetic IoT Services	Simulated service datasets with injected QoS fluctuations
04	Synthetic IoT Services	Synthetic datasets with spatial distributions
05	114 Primary Studies	Systematic review - papers from IEEE, ACM, Scopus, WoS
06	RC-MPSP Benchmarks	Standard resource-constrained multi-project scheduling benchmarks

Paper	Dataset	Description
07	Synthetic IoT Scenarios	Generated service composition scenarios
08	Synthetic IoT Services	Simulated service scenarios with user preferences
09	Synthetic IoT Services	Simulated IoT service datasets
10	Market Simulation	Simulated market with provider and user agents
11	300+ Primary Studies	Survey - papers from IEEE, ACM, Springer, Elsevier
12	Crowdsourced Simulation	Simulated environment with user providers
13	CEaaS Simulation	Simulated energy marketplace
14	Variable Renewable Simulation	Simulated intermittent energy patterns
15	Mobile IoT Simulation	Simulated mobile device trajectories
16	Synthetic IoT Prosumer Simulation	Simulated crowdsourced energy pool scenarios
17	Synthetic Moving IoT Services	Simulated mobility patterns (random waypoint, vehicle, drone)

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